FEDERAL SPECIFICATION

TIN FOIL (FOR DENTAL USE)

This specification was approved on the above date by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers tin foil suitable for use in the dental laboratory as a burnished covering for models in the construction of dentures in order that a smooth, accurate surface may be imparted to the finished product.

1.2 Classification.

1.2.1 *Type and sizes.*—Tin foil covered by this specification shall be of one type and the following sizes, as specified in the invitation for bids:

Size 1.—One-pound roll
Size 2.—Half-pound roll

1.2.2 *Thicknesses.*—Tin foil covered by this specification shall be of the following thicknesses, as specified in the invitation for bids:

0.001 inch (see 3.3)
0.002 inch (see 3.3)
0.003 inch (see 3.3)

2. APPLICABLE SPECIFICATIONS

2.1 There are no other specifications applicable to this specification except those referenced in section 7 by individual departments.

(Copies of Federal specifications and the Federal Specifications Index may be obtained upon application, accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.)

3. REQUIREMENTS

3.1 Material.—The tin foil covered by this specification shall be that known as commercial grade A, containing not less than 99.80 percent pure tin.

3.2 Sizes.—Tin foil shall be furnished in 1-pound or ½-pound rolls, as specified in the invitation for bids.

3.3 Dimensions.—Tin foil shall be approximately 6 inches wide and shall be of the following thicknesses as specified in the invitation for bids: 0.001 inch, 0.002 inch, and 0.003 inch with a plus tolerance of 0.0003 inch applicable to each thickness.

3.4 Workmanship.—Tin foil shall be free from breaks, holes, and other imperfections adversely affecting its suitability for the purpose intended.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling.—Not less than three rolls of tinfoil shall be taken at random from each delivery for inspection and tests.

4.2 Inspection.—Tin foil shall be inspected for workmanship, material, weight of rolls, dimensions, packaging, packing, and marking.

4.3 Tests.

4.3.1 *Determination of percentage of tin.*—Specimens shall be taken from each of the

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rolls and the following analysis shall be made: Weigh accurately about 0.5 gram of tin foil and dissolve in 25 milliliters of concentrated HCl by heating in a 500-milliliter covered beaker. Dilute the solution to 100 milliliters with cold water, add methyl red indicator, and neutralize the solution with ammonia. Add about 50 milliliters of saturated solution of ammonium nitrate and dilute the solution to 300 milliliters with boiling water, stirring the contents constantly. Allow the beaker to stand on the steam bath until the precipitate has settled. Decant the supernatant liquor through a hard filter paper and wash the precipitate six times by decantation, using 100 milliliters of boiling 1-percent HNO_3 and allow the precipitate to settle after each washing. After the precipitate has been completely transferred to the filter paper, thoroughly wash and allow to drain; transfer to a porcelain crucible; dry on a hot plate; ignite at a low temperature until the filter paper has been consumed; and finally dry to constant weight.

$$\frac{\text{Weight of } SnO_2 \times 0.78766 \times 100}{\text{Weight of sample in grams}} = \text{percent tin}$$

5. PREPARATION FOR DELIVERY

5.1 Packaging. — Unless otherwise specified, each roll of tin foil shall be packaged in accordance with accepted commercial practice.

5.2 Packing. — The subject commodity shall be prepared for shipment to permit acceptance by carrier for transportation at the lowest applicable rate and to afford maximum protection from normal hazards of transportation.

5.3 Marking.

5.3.1 *Packages.* — Unless otherwise specified, packages shall be marked with the name, size, and thickness of the commodity, the quantity contained therein, and the names of the manufacturer and contractor.

5.3.2 *Shipping containers.* — Unless otherwise specified, shipping containers shall be marked with the name, size, and thickness of the commodity as defined by the contract or order under which the shipment is made, quantity contained therein, name of the contractor, number of the contract or order, name and address of the consignee, and the gross weight.

6. NOTES

6.1 Ordering data. — Purchasers should specify size and thickness required and should exercise any desired options offered herein. (See 1.2.1, 1.2.2, 3.2, 3.3, 5.1, 5.3, 6.2, and 7.2.6.)

6.2 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.3 This specification covers the following item appearing in the Armed Services Catalog of Medical Matériel:

5–312–050—Foil, Tin, ½ pound.

6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the type, etc., normally obtained for Federal Government requirements for the purpose specified.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related

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Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

7. DEPARTMENTAL REQUIREMENTS

7.1 The following specifications, of the issues in effect on date of invitation for bids, and special requirements, form a part of this specification for purchases made under this specification by the respective departments.

7.2 **Army, Navy, and Air Force.**

7.2.1 *Applicable specifications:*

Federal Specifications:

NN-B-601—Boxes, Wood-Cleated-Plywood, for Domestic Shipment.

NN-B-621—Boxes, Wood, Nailed and Lock-Corner.

NN-B-631—Boxes; Wood, Wirebound (for Domestic Shipment).

Military Specifications:

JAN-P-105—Packaging and Packing for Overseas Shipment — Boxes, Wood, Cleated, Plywood.

JAN-P-106—Packaging and Packing for Overseas Shipment — Boxes, Wood, Nailed.

JAN-B-107—Boxes, Wood, Wirebound (Overseas Type).

JAN-P-120—Packaging and Packing for Overseas Shipment — Cartons, Folding, Paperboard.

JAN-P-125—Packaging and Packing for Overseas Shipment—Barrier-Materials, Waterproof, Flexible.

JAN-P-133—Packaging and Packing for Overseas Shipment — Boxes, Set-Up, Paperboard.

JAN-P-140—Packaging and Packing for Overseas Shipment — Adhesives, Water-Resistant, Case-Liner.

MIL-M-3157—Marking of Domestic Shipments by Contractors for Supplies and Equipment Consigned to the Medical Departments of the Armed Services.

Navy Department Specification:

General Specifications for Inspection of Material.[1]

(Copies of specifications required for Army and Air Force purchases may be obtained upon application to the procuring agency. Both the title and identifying symbol or number should be stipulated when requesting copies.

(Copies of Federal and Military specifications required for Navy purchases and Navy Department specifications may be obtained without cost upon application to the Bureau of Supplies and Accounts, Department of the Navy, Washington 25, D. C., except that Naval activities should make application to the Commanding Officer, Naval Supply Depot, Scotia 2, New York. Both the title and identifying symbol or number should be stipulated when requesting copies of specifications.)

7.2.2 *Inspection procedures (Navy only).*— The general inspection procedures shall be in accordance with Navy Department Specification, General Specifications for Inspection of Material.

7.2.3 *Packaging.*

7.2.3.1 *Unit packaging.* — Each ½-pound roll of tin foil as specified constitutes one unit. Each unit shall be evenly wound on a core or spool of appropriate size. Each unit shall be completely wrapped in tissue paper and packaged in a carton of appropriate size, conforming to the requirements of Military Specification JAN-P-120 or JAN-P-133. Each carton shall be adequately secured to prevent accidental opening.

7.2.3.2 *Intermediate packaging.*—Six unit packages (6 units) shall be packaged in a carton of appropriate size and design or shall be completely wrapped with kraft paper of

[1] Applicable to Navy purchases only.

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suitable weight. Carton or wrapping shall be adequately secured to prevent accidental opening.

7.2.4 *Packing.*

7.2.4.1 *Domestic.* — Sixteen intermediate packages (96 units) shall be packed in a shipping container conforming to the requirements of Federal Specification NN-B-601, NN-B-621, or NN-B-631.

7.2.4.2 *Overseas.* — Sixteen intermediate packages (96 units) shall be packed in a shipping container designed for type 2 load, and conforming to the requirements of Military Specification JAN-P-105, JAN-P-106, or JAN-B-107. Closure and strapping shall conform to the requirements of appendix to applicable box specification. Each shipping container shall be lined with a sealed waterproof bag, or its equivalent, made from material conforming to the requirements of Military Specification JAN-P-125 for case liners; seams and closures shall be completely sealed with a continuous seam at least ¾ inch wide of water-resistant adhesive conforming to the requirements of Military Specification JAN-P-140.

7.2.4.3 *Packing variation permitted.* — If the number of units in any shipment is not the exact multiple of the number of units specified to be contained in each shipping container, the units not constituting the nearest such multiple shall be packed in a shipping container of suitable size and design, so constructed as to insure safe delivery to specified destination.

7.2.5 *Marking.*

7.2.5.1 *Packages.*—Each unit package and each intermediate package shall be marked with the stock number, nomenclature, unit, quantity, and the name of the contractor. When the contractor is not the manufacturer, the name of the manufacturer shall also appear as the manufacturer.

7.2.5.2 *Shipping containers.*

7.2.5.2.1 *Domestic.* — Each shipping container shall be marked in accordance with Military Specification MIL-M-3157.

7.2.5.2.2 *Overseas.* — Each shipping container shall be marked as specified by the procuring agency.

7.2.6 *Ordering data.* — Procurement documents should specify whether the subject commodity is to be packed, and shipping containers marked for domestic or overseas shipment (see 7.2.4 and 7.2.5.2.).

NAVY INTEREST: Md

This specification is a part of Section IV, Part 5, of the Federal Standard Stock Catalog.

For sale by the Superintendent of Documents, Washington 25, D. C. Price 5 cents.

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